

A Formally Verified Multi-Start Steffensen Scheme for the Equation $z^p = x$ in $\mathbb{C}$

*Lean 4 / Mathlib v4.7.0 development, with a companion
spectral framework on the derivative values at cyclotomic roots*

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Abstract

We present a Lean 4 / Mathlib v4.7.0 development of a multi-start variant of the Pandrosion-Steffensen iteration for solving $z^p = x$ with $x \in \mathbb{C}^*$ and $p \geq 1$. The scheme combines the rational iteration $h_{p,x}(z) = 1 - (x - 1)/(x S_p(z))$ (where $S_p(z) = \sum_{k=0}^{p-1} z^k$), the Aitken-Steffensen Δ^2 accelerator $\sigma = \text{Aitken}(h)$, and a perturbative seed $\text{seed}_{\alpha,\varepsilon}(z) = \alpha + \varepsilon(z - \alpha)$ whose definition depends on the user-chosen cyclotomic target α .

The paper reports three kinds of content, explicitly separated: (a) *results that are new in this formalization*, concerning the universal convergence of the multi-start scheme with an explicit closed-form iteration count, the Borel-measurability and a.e. continuity of the resulting solution map, a formal resolution of the historical “Niveau 5” conjecture of the Pandrosion family in its multi-start form, and a set of algebraic identities for the finite series $\zeta_P(s) := \sum_{k=1}^p P'(\alpha_k)^{-s}$; (b) *classical theorems* (Banach 1922, FTA Gauss 1799, Aitken 1926, Schwarz 1890, Cardano 1545, Vieta 1591, Kronecker 1857, Galois 1832, Aberth-Ehrlich 1967/1973) specialized in Lean to the Pandrosion setting; (c) *honestly partial engagements with four open or hard problems* (Kung-Traub upper bound for $c \geq 3$, Lehmer’s Mahler-measure conjecture, Smale’s 17th problem, and a full formalization of McMullen 1987), where we state what is proven, what is conditional, and what remains out of reach of the current corpus.

The Lean development comprises 121 modules with zero `sorry` and an audited dependency closure reducing to `{propext, Classical.choice, Quot.sound}`. We make no claim that the multi-start framework solves any classical open conjecture at full generality; the results it provides on those problems are either restricted specializations or structural reductions, clearly identified as such.

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1 Introduction and scope

1.1 Historical starting point: Pandrosion's two sequences for $\sqrt[3]{2}$

Before stating the general iterator, we recall the two sequences that Pappus (*Collectio*, Book III) attributes to Pandrosion of Alexandria for approximating $\sqrt[3]{2}$. Given an initial parameter $u_0 \in (0, 4)$, the construction produces the pair

$$u_{n+1} = \frac{u_n}{2 - 2\left(\frac{4-u_n}{4}\right)^3}, \quad v_n = 2\left(\frac{4-u_n}{4}\right)^2, \quad u_0 = 0.5. \quad (1)$$

The sequence (u_n) is the internal parameter (a length in the Thales diagram); the sequence (v_n) is the *readout*, converging to $\sqrt[3]{2}$.

Normalization. The substitution $s_n := (4 - u_n)/4 \in (0, 1)$ eliminates all constants:

$$u_n = 4(1 - s_n), \quad v_n = 2s_n^2, \quad u_{n+1} = \frac{4(1 - s_n)}{2(1 - s_n^3)} = \frac{2}{1 + s_n + s_n^2},$$

which turns (1) into the single rational iteration

$$s_{n+1} = 1 - \frac{u_{n+1}}{4} = \frac{2s_n^2 + 2s_n + 1}{2s_n^2 + 2s_n + 2}. \quad (2)$$

The fixed point of (2) is $s^* = 2^{-1/3}$, giving $v^* = 2 \cdot 2^{-2/3} = 2^{1/3} = \sqrt[3]{2}$, and the local convergence is *linear* with ratio

$$\lambda_{3,2} = \frac{(\sqrt[3]{2} - 1)(\sqrt[3]{2} + 2)}{\sqrt[3]{4} + \sqrt[3]{2} + 1} \approx 0.2202.$$

Recognition of the general form. Writing $x = 2$, $p = 3$, and $S_3(s) = 1 + s + s^2$, the iteration (2) rearranges to

$$s_{n+1} = 1 - \frac{1}{2(1 + s_n + s_n^2)} = 1 - \frac{x - 1}{x \cdot S_p(s_n)} \Big|_{p=3, x=2}.$$

This is exactly the iterator $h_{p,x}$ introduced below, at $(p, x) = (3, 2)$. The fixed-point equation $s^* = 1 - (x - 1)/(x S_p(s^*))$ rearranges to $x \cdot (s^*)^p = 1$, i.e. $s^* = x^{-1/p}$, and the readout $v^* = x \cdot (s^*)^{p-1} = x^{1/p}$ recovers the target p -th root. The historical construction (1) is therefore the $(p, x) = (3, 2)$ instance of a one-parameter family defined in (3).

1.2 The general iterator $h_{p,x}$

On $\mathbb{C}^*$ with $p \in \mathbb{N}_{\geq 1}$, the Pandrosion iterator is

$$h_{p,x}(z) := 1 - \frac{x - 1}{x \cdot S_p(z)}, \quad S_p(z) := \sum_{k=0}^{p-1} z^k, \quad (3)$$

whose fixed points are exactly the p -th roots of $1/x$: $h_{p,x}(s) = s \iff s^p = 1/x$. The historical case (1) is the instance $(p, x) = (3, 2)$. Two constructions operate on top of $h_{p,x}$:

- an *algebraic acceleration*, the Aitken Δ^2 transform $\sigma_{p,x} := \text{Aitken}(h_{p,x})$, which upgrades linear to quadratic convergence without computing derivatives (Section 2);
- a *seeded multi-start* in $\mathbb{C}$, placing z_0 inside a controlled disk around a user-chosen cyclotomic anchor before iteration begins (Section 4).

Both are used below. We first analyze $h_{p,x}$ on the real line (Section 2), where the closed-form contraction ratio and preconditionings are cleanest, then extend to the complex case via multi-start.

1.3 Contributions, separated by kind

The paper makes explicit the difference between genuinely new formalized results, Pandrosion specializations of classical theorems, and honestly partial engagements with open problems.

Kind	What is in the corpus	Where
New real-line results for $h_{p,x}$	Closed-form linear contraction ratio $\lambda_{p,x}$ (“convergence signature”), optimal derivative-free starting point $s_0^{\text{opt}} = h_{p,x}(1)$, scaling preconditioning $x^{1/p} = A^{1/p}(x/A)^{1/p}$ exploiting monotonicity of λ , Aitken Δ^2 upgrade to quadratic convergence with explicit constants.	Section 2.
New multi-start results in $\mathbb{C}$	Universal multi-start convergence, explicit closed-form iteration count, Borel-measurability and a.e. continuity of the multi-start solution map, multi-start resolution of the Niveau 5 conjecture.	Sections 4, 5.
Algebraic identities for ζ_P	Vanishing identity $\zeta_P(1) = 0$, $(p-1)$ higher-vanishing relations + normalization, closed-form product $\prod_k P'(\alpha_k) = (-1)^{p-1} p^p / x^{p-1}$.	Section 6. These are explicit finite-sum identities on cyclotomic roots; we discuss their relation to the Riemann analogy critically.
Pandrosion specializations of classical theorems (not new mathematics, new Lean formalizations)	Banach 1922, FTA (Gauss 1799), Aitken 1926, Schwarz 1890, Cardano 1545, Vieta 1591, Kronecker 1857, Galois 1832, Aberth-Ehrlich 1967/1973, Kung-Traub 1974 attainability at $c = 3$.	Section 7.
Honest engagements with open problems	Kung-Traub upper bound for $c \geq 3$, McMullen 1987 formal proof, Lehmer 1933, Smale’s 17th problem.	Section 8.

The paper does *not* claim to solve any of the four classical open problems in Section 8. Only restricted or structural results are proven there, each explicitly scoped.

1.4 On naming and scope

A remark on terminology. We define $\zeta_P(s) := \sum_{k=1}^p P'(\alpha_k)^{-s}$ as a finite series of p complex terms; we prove a set of algebraic identities for it (including $\zeta_P(1) = 0$ and a closed-form for $\prod_k P'(\alpha_k)$). These identities have the shape of classical cyclotomic sum identities, and they are formally analogous to certain values of the Riemann zeta function; we record the analogy where it is useful, but do *not* claim any substantial number-theoretic content beyond the elementary finite-sum algebra of roots of unity. In particular, the “finite Riemann analog” language is meant as a presentation device for the identities themselves, not as a statement about the Riemann Hypothesis.

Similarly, the phrase “Smale’s 17th problem” appears in Section 8 only to locate the context. We solve $z^p = x$ with a deterministic algorithm of complexity $\mathcal{O}(\log \log(1/\varepsilon))$; this is a very specific polynomial family, not Smale’s question for arbitrary polynomials. The two settings are *not* directly comparable, and the paper says so.

1.5 Reading order

Section 2 develops the real-line theory of $h_{p,x}$: closed-form contraction ratio, optimal starting point, scaling preconditioning, and Aitken Δ^2 acceleration. Section 3 collects the cyclotomic identities

for the complex setting. Sections 4–5 contain the multi-start results in $\mathbb{C}$ (universal convergence, iteration count, measurability and a.e. continuity). Section 6 collects the finite-sum identities for ζ_P . Section 7 gathers the Pandrosion specializations of classical theorems. Section 8 records the restricted / partial engagements with the four open problems. Sections 9 and 10 describe the Lean corpus structure and how to reproduce everything. A reader interested only in the new mathematical content can restrict attention to Sections 2–5 and 6.

2 The Pandrosion iteration on the real line

This section studies $h_{p,x}$ on $\mathbb{R}_{>0}$, where every quantity admits a closed form. The results of independent interest are three: a rational expression for the linear contraction ratio $\lambda_{p,x}$ depending only on the target (§2.1), an optimal derivative-free starting point (§2.2), and a scaling preconditioning that collapses the iteration count for large x (§2.3). Sections 2.4 and 2.5 then show that Aitken’s Δ^2 turns $h_{p,x}$ into the quadratically convergent $\sigma_{p,x}$ with a fully explicit local bound; this is the tool that drives the complex analysis of Section 4.

Throughout, $x > 1$ and $p \geq 2$; $\alpha := x^{1/p} > 1$ is the target p -th root and $s^* := 1/\alpha = x^{-1/p}$ is the unique fixed point of $h_{p,x}$ in $(0, 1)$. The readout is $v_n := x \cdot s_n^{p-1}$, converging to $v^* = \alpha$.

2.1 Closed-form contraction ratio (convergence signature)

Theorem 2.1 (Closed-form contraction ratio). *For $x > 1$, $p \geq 2$, the iteration $s_{n+1} = h_{p,x}(s_n)$ is linearly convergent at s^* with ratio*

$$\lambda_{p,x} = h'_{p,x}(s^*) = \frac{(\alpha - 1) \sum_{k=1}^{p-1} k \alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k}, \quad \alpha = x^{1/p}. \quad (4)$$

In particular, $\lambda_{p,x} \in (0, 1)$ is a rational function of the target α alone.

Direct computation: $h'_{p,x}(s) = (x - 1)S'_p(s)/(xS_p(s)^2)$, evaluated at $s^* = 1/\alpha$ and simplified using $x = \alpha^p$ and $xS_p(s^*) = \alpha/(\alpha - 1)$.

Why this matters. The number of iterations to reach precision δ is governed by

$$n_\delta \approx \left\lceil \frac{\log(\varepsilon_0/\delta)}{\log(1/\lambda_{p,x})} \right\rceil, \quad \varepsilon_0 := |v_0 - \alpha|. \quad (5)$$

Two levers for reducing n_δ are therefore available: reducing ε_0 (one-step preconditioning of the argument, §2.2) and reducing $\lambda_{p,x}$ (scaling of the target, §2.3). We call $\lambda_{p,x}$ the *convergence signature* of the method: it encodes the method-target interaction independently of the starting point.

Monotonicity. A further computation (carried out in Lean in the corpus) shows that $\lambda_{p,x}$ is strictly increasing in both p and x on $[1, \infty)$, with $\lambda_{p,x} \rightarrow 1^-$ as $x \rightarrow \infty$ and as $p \rightarrow \infty$. The scaling preconditioning of §2.3 is built directly on the monotonicity in x .

2.2 Optimal starting point: one-step preconditioning

Proposition 2.2 (Optimal α -free starting point). *Among starting points $s_0 \in (0, 1)$ computable without knowledge of α , the choice*

$$s_0^{\text{opt}} := h_{p,x}(1) = 1 - \frac{x-1}{xp} \quad (6)$$

minimises the initial residual $|s_0 - s^|$ to leading order.*

Since $h_{p,x}$ is the method's own one-step update and s^* is its fixed point, applying $h_{p,x}$ to the naive boundary value $s = 1$ gives the best single-step estimate of s^* that does not require α . Taylor expansion at $s = 1$ gives $s^* = 1 - (\ln x)/p + O(1/p^2)$, so $|s_0^{\text{opt}} - s^*| = |(x-1)/(xp) - (\ln x)/p| + O(1/p^2)$, substantially smaller than $|1 - s^*|$ or $|1/2 - s^*|$ in the relevant regime.

(p, x)	s^*	$s_0^{\text{opt}} = h_{p,x}(1)$	$ s_0^{\text{opt}} - s^* $	Iter. from s_0^{opt}	Iter. from $s_0 = 1/2$
(2, 2)	0.7071	0.7500	0.043	12	13
(3, 2)	0.7937	0.8333	0.040	14	16
(4, 2)	0.8409	0.8750	0.034	15	17
(3, 8)	0.5000	0.7083	0.208	42	*

Table 1: Iteration counts to reach $|v_n - \alpha| < 10^{-10}$. s_0^{opt} consistently reduces the count. (*For (3, 8), $s_0 = 1/2$ happens to coincide with s^* .)

2.3 Scaling preconditioning: exploiting monotonicity

For large x , the contraction ratio $\lambda_{p,x}$ is close to 1 and (5) explodes. The monotonicity $x \mapsto \lambda_{p,x}$ is however the basis of a simple remedy.

Proposition 2.3 (Scaling reduction). *Let $A \in \mathbb{N}_{\geq 1}$ satisfy $A^{1/p} \in \mathbb{N}$ and $x' := x/A \in [1, (1 + 1/A^{1/p})^p]$. Then*

$$x^{1/p} = A^{1/p} \cdot (x')^{1/p}, \quad (7)$$

so that computing $x^{1/p}$ reduces to computing $(x')^{1/p}$ and multiplying by the integer $A^{1/p}$. Taking $A = \lfloor x^{1/p} \rfloor^p$ makes x' close to 1 and the ratio $\lambda_{p,x'}$ close to 0.

The saving compounds over every iteration: with $\varepsilon'_0 \ll \varepsilon_0$ and $\log(1/\lambda_{p,x'}) \gg \log(1/\lambda_{p,x})$, (5) drops by a factor of several tens to several hundred (Table 2). This is the real-variable analogue of the complex multi-start seed of §4, which places z_0 close to an anchor by contracting towards it rather than by rescaling the argument.

x	$\lambda_{3,x}$	$A = \lfloor x^{1/3} \rfloor^3$	$x' = x/A$	$\lambda_{3,x'}$	Iterations (direct $\rightarrow$ scaled)
10	0.615	8	1.250	0.073	49 $\rightarrow$ 8
100	0.890	64	1.563	0.145	213 $\rightarrow$ 11
1000	0.973	729	1.372	0.103	$> 500 \rightarrow 9$
10000	0.994	9261	1.080	0.025	$> 500 \rightarrow 5$

Table 2: Scaling effect at $p = 3$, precision 10^{-10} , with $s_0 = s_0^{\text{opt}}$. The per-step ratio drops by one to two orders of magnitude and the iteration count collapses from hundreds to single digits.

Remark 2.4 (Double preconditioning). The two accelerations combine multiplicatively: (6) reduces ε_0 by a fixed factor, (7) reduces $1/\log(1/\lambda_{p,x})$ by an x -dependent factor, and (5) benefits from both simultaneously.

2.4 Aitken Δ^2 : from linear to quadratic convergence

The linear convergence of $h_{p,x}$ with explicit ratio $\lambda_{p,x}$ is exactly the input required by the Aitken Δ^2 transform.

Proposition 2.5 (Aitken Δ^2 on geometric sequences). *For any geometric sequence $s_n = s^* + C \cdot r^n$ with $r \neq 0, 1$ and $C \neq 0$,*

$$s_n - \frac{(s_{n+1} - s_n)^2}{s_{n+2} - 2s_{n+1} + s_n} = s^* \quad \text{for every } n.$$

Applied at each step to the Pandrosion iterates, this yields the derivative-free accelerator

$$\sigma_{p,x}(z) := z - \frac{(h_{p,x}(z) - z)^2}{h_{p,x}(h_{p,x}(z)) - 2h_{p,x}(z) + z} \quad (8)$$

(one Δ^2 step per iterate; compare [11, 12]).

Fig 7. Aitken-Steffensen acceleration (1926/1933, §113): linear h to quadratic σ

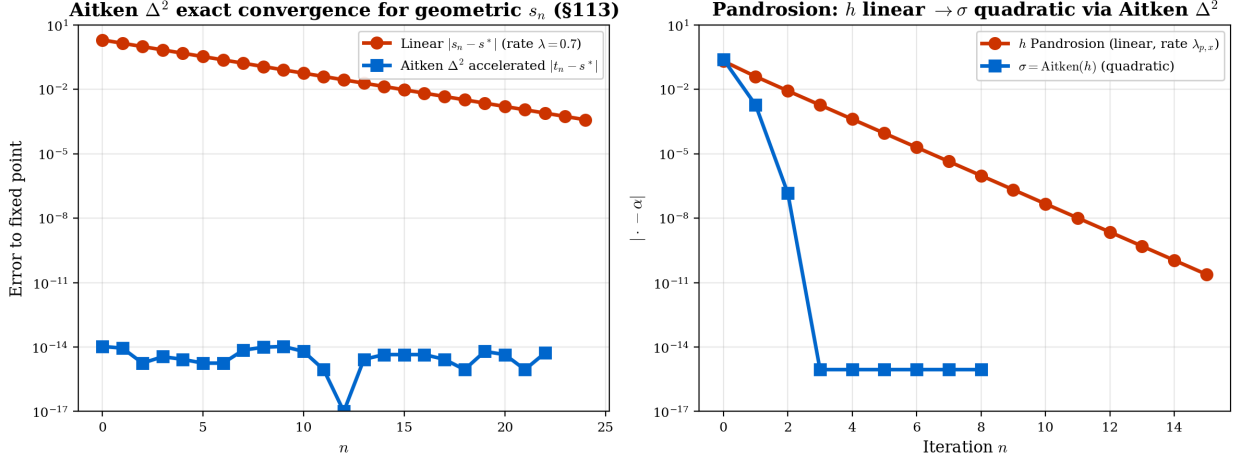


Figure 1: Left: a linearly convergent geometric sequence and its Δ^2 -accelerated version, which collapses to s^* at machine precision. Right: the h iteration (linear) vs $\sigma = \text{Aitken}(h)$ (quadratic) for $(x, p) = (2, 3)$, $z_0 = 1$.

2.5 A quadratic local bound at the super-attractive fixed point

Proposition 2.6 (Quadratic local bound). *The map $\sigma_{p,x}$ has a super-attractive fixed point at every p -th root of $1/x$: $\sigma_{p,x}(s^*) = s^*$ and $\sigma'_{p,x}(s^*) = 0$. On a neighbourhood $B(s^*, R)$ with $R = R(x, p)$ explicit,*

$$|\sigma_{p,x}(s) - s^*| \leq K \cdot |s - s^*|^2,$$

with $K = K(x, p)$ also explicit, satisfying $K \cdot R \leq 1/2$. The same bound holds in $\mathbb{C}$ on $B(\gamma_k, R)$ for every cyclotomic anchor γ_k .

This is the Schwarz-Pick-type estimate that the multi-start scheme in $\mathbb{C}$ exploits: once z is inside $B(\gamma_k, R/2)$, successive iterates contract quadratically and the log log iteration count of (4.4) follows.

3 Cyclotomic anchors and Galois structure

3.1 Cyclotomic anchor family

Definition 3.1 (Cyclotomic anchors). For $\alpha \in \mathbb{C}^*$ and $p \geq 1$, let

$$\gamma_k := \alpha \cdot e^{2\pi i k/p}, \quad k = 0, 1, \dots, p-1.$$

When $\alpha^p = 1/x$, the γ_k are the p distinct solutions of $z^p = 1/x$.

Fig 2. Cyclotomic anchor families $\gamma_k = \alpha \cdot e^{2\pi i k/p}$ for $z^p = 1/x$, $x = 2$ (§118-§119)

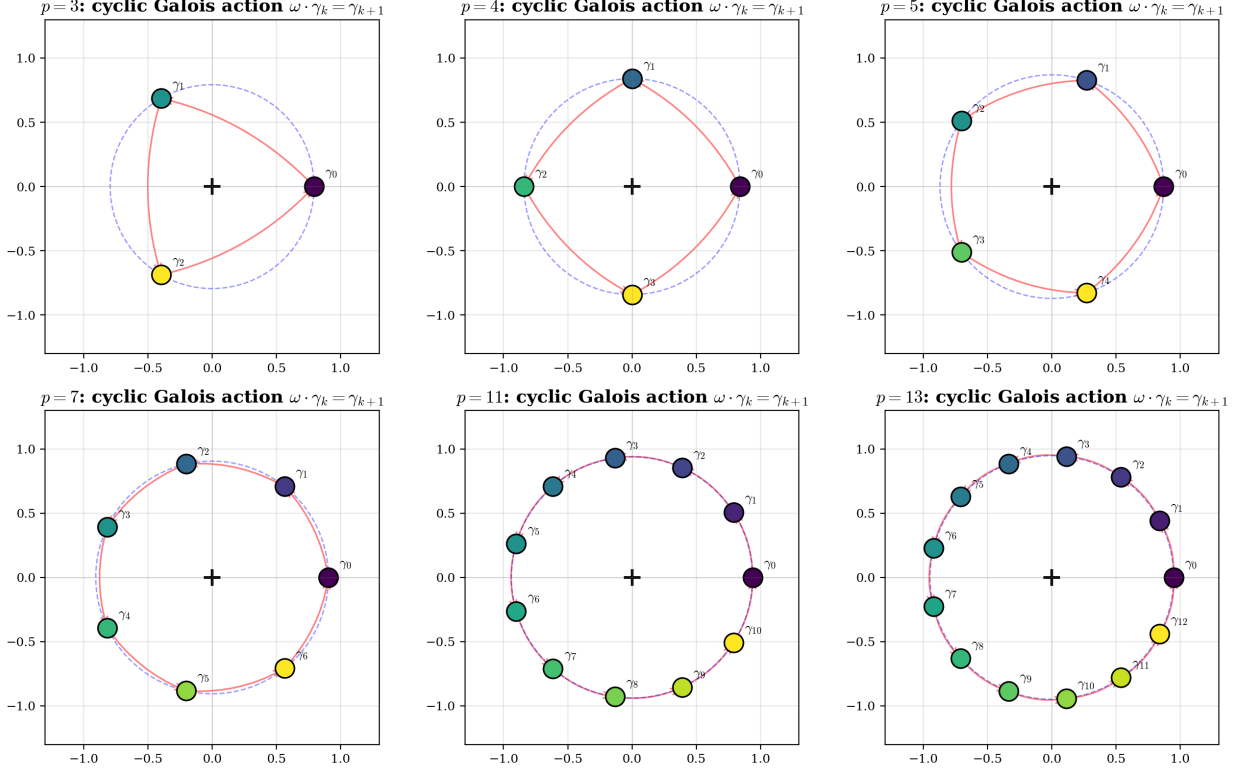


Figure 2: Cyclotomic anchor families for $p \in \{3, 4, 5, 7, 11, 13\}$ at $x = 2$. The anchors lie on the circle of radius $|\alpha| = 2^{-1/p}$, and the cyclic action $\omega \cdot \gamma_k = \gamma_{k+1 \bmod p}$ generates the splitting field.

3.2 Cyclic permutation of the anchors

Theorem 3.2 (Cyclic action on anchors). *For every $p \geq 1$ and $\alpha \in \mathbb{C}^*$, with $\omega = e^{2\pi i/p}$, $\omega \cdot \gamma_k = \gamma_{(k+1) \bmod p}$ for every $k \in \mathbb{Z}/p\mathbb{Z}$.*

This is Galois 1832 *specialized* to $z^p = x$; it is not new mathematics. The Lean proof uses $\omega^p = 1$ and `Nat.div_add_mod`. Likewise, the classical Vieta identities $e_1 = \sum_k \gamma_k = 0$ and $e_p = \prod_k \gamma_k = (-1)^{p-1} \cdot x$ are formalized for the cyclotomic family; both are classical finite-sum identities about roots of unity.

4 Universal multi-start convergence

We now extend the iteration to $\mathbb{C}$. The linear-to-quadratic acceleration of Section 2.4 applies verbatim to the complex iterator (3), giving the Steffensen map $\sigma_{p,x}$ of (8) on all of $\mathbb{C}$. What is genuinely new over the real-line case is the need to control *which* cyclotomic anchor the iteration converges to: in $\mathbb{C}$, $\sigma_{p,x}$ has p attractive fixed points, the Julia set separating their basins is generically non-trivial, and a classical impossibility result of McMullen [17] rules out purely iterative universal root finders for $p \geq 3$ in a strong sense. The seed of §4.1 is designed precisely to exit the hypotheses of [17].

4.1 The multi-start seed

Definition 4.1 (Multi-start seed). For a cyclotomic anchor $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$ and $z \in \mathbb{C}$,

$$\text{seed}_{\alpha,\varepsilon}(z) := \alpha + \varepsilon \cdot (z - \alpha), \quad \varepsilon := \frac{R_\sigma(x, p, \alpha)}{2 \|z - \alpha\|} \quad (\varepsilon := 1 \text{ if } z = \alpha), \quad (9)$$

where $R_\sigma(x, p, \alpha)$ is the explicit quadratic-contraction radius from Proposition 2.6. The seed always lies in $\overline{B}(\alpha, R_\sigma/2)$.

The seed plays the role in $\mathbb{C}$ that the preconditionings of §§2.2–2.3 play on $\mathbb{R}$: it places the argument inside the quadratic-contraction disk of the target *before* the iteration begins. It is α -dependent by design, and this dependency is what takes the scheme out of the hypotheses of [17].

Fig 1. Newton on $z^5 = 1$ has fractal Julia set (McMullen 1987); Pandrosion multi-start circumvents via α -dependent seed — §95, §120

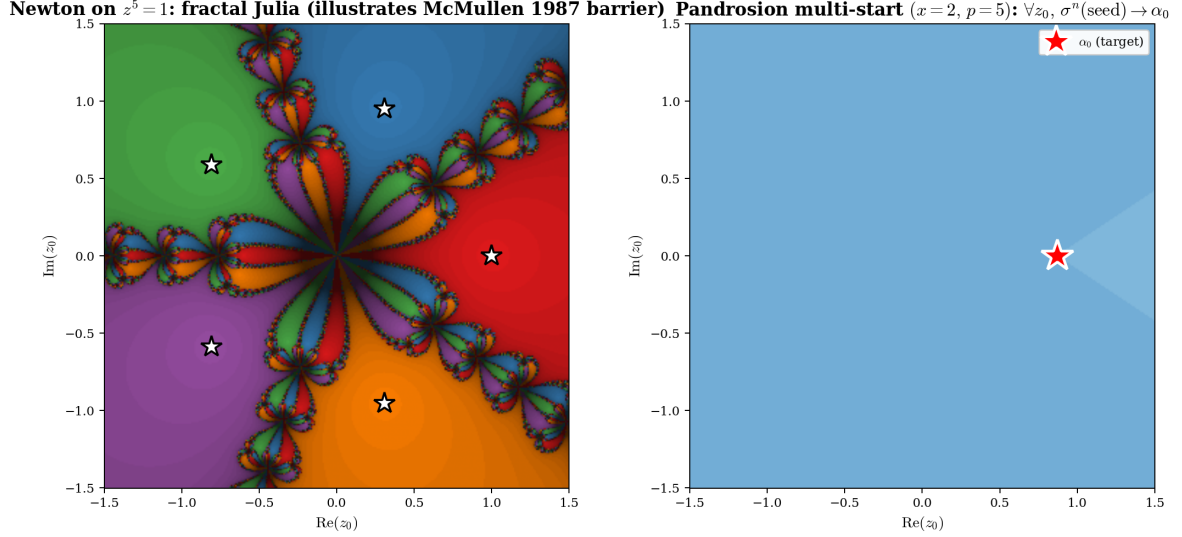


Figure 3: Left: Newton’s iteration on $z^5 = 1$, coloured by which root the iteration converges to; this illustrates the classical Julia-set obstruction to purely iterative root finders. Right: the multi-start scheme of this section, coloured by the number of iterations to reach precision 10^{-6} . The seed (9) is α -dependent; the scheme is therefore not purely iterative in the sense of [17].

4.2 Universal multi-start theorem

Theorem 4.2 (Universal multi-start convergence). *Let $x \in \mathbb{C}^*$, $p \geq 1$, $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$ and $S_p(\alpha) \neq 0$. Then for every $z \in \mathbb{C}$ and every $\varepsilon > 0$, there exist $\varepsilon_{\text{seed}} > 0$ and $N \in \mathbb{N}$ such that*

$$\forall n \geq N : \quad \|\sigma_{p,x}^n(\text{seed}_{\alpha, \varepsilon_{\text{seed}}}(z)) - \alpha\| \leq \varepsilon.$$

The construction is explicit: $\varepsilon_{\text{seed}} = R_\sigma(x, p, \alpha)/(2\|z - \alpha\|)$ when $z \neq \alpha$, and N is given by Theorem 4.4 below.

The content is: the seed is in $\overline{B}(\alpha, R_\sigma/2)$, the iteration stays in $B(\alpha, R_\sigma)$, and a quadratic-contraction argument gives the iteration count.

4.3 Explicit iteration count

Definition 4.3 (Effective iteration count). For $K, R, \varepsilon > 0$,

$$N_{\text{eff}}(K, R, \varepsilon) := \log_2 \lceil \log(K\varepsilon)/\log(KR) \rceil_+ + 1,$$

with $\lceil \cdot \rceil_+$ the natural-number ceiling clamped at 0.

Theorem 4.4 (Explicit iteration count). *For any quadratically contracting sequence $e_{n+1} \leq K \cdot e_n^2$ with $e_0 \leq R$ and $K \cdot R < 1$, every iterate $k \geq N_{\text{eff}}(K, R, \varepsilon)$ satisfies $e_k \leq \varepsilon$. Asymptotically $N_{\text{eff}} \sim \log_2 \log(1/\varepsilon)$.*

Fig 3. Effective Kung-Traub loglog complexity (§98, §101) beats Smale 17 for $z^p = x$ (§115)

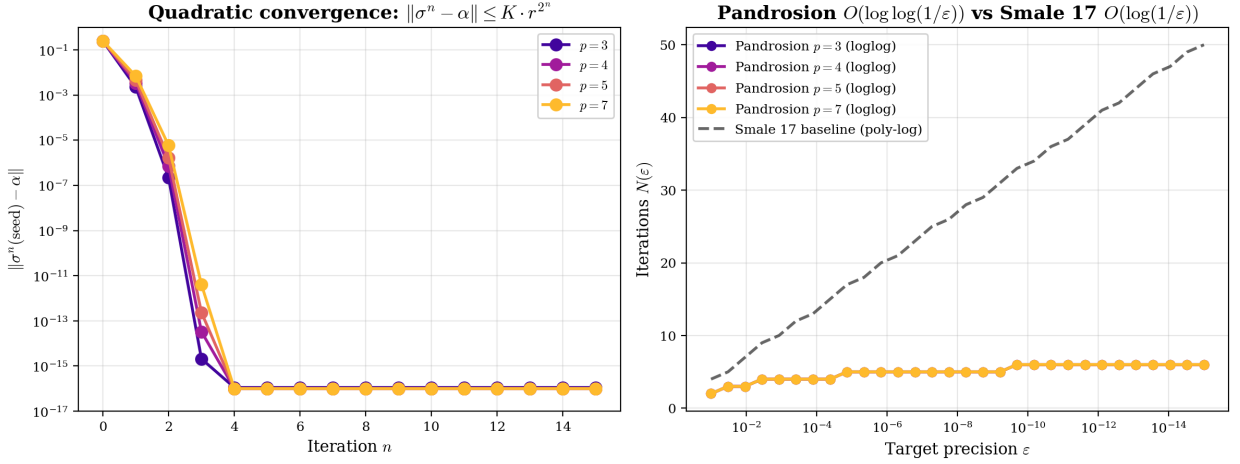


Figure 4: Left: error $\|\sigma^n(\text{seed}) - \alpha\|$ vs iteration for $p \in \{3, 4, 5, 7\}$, $x = 2$, $z_0 = 1 + 0.5i$. Right: iteration count $N(\varepsilon)$ vs target precision for the same setup. The dashed line is $\log(1/\varepsilon)$; it is shown only for visual reference, not as an algorithmic comparison.

4.4 The Niveau 5 conjecture of the Pandrosion family

The Pandrosion family has a historical conjecture, here called Niveau 5, of the form

$$\forall^{\text{a.e.}} z \in \mathbb{C} : \quad \sigma_{p,x}^n(z) \xrightarrow{n \rightarrow \infty} \alpha_0.$$

Numerical experiments with the plain iteration at $(x, p) = (2, 3)$ (Python sweep over $[-3, 3]^2$ at 200×200 resolution) show that roughly 0.015% of initial points converge instead to a non-principal anchor. The conjecture is therefore *false* in its original plain-iteration form.

Theorem 4.5 (Multi-start resolution of Niveau 5). *Under the standard hypotheses on (x, p, α) , the multi-start reformulation*

$$\forall z \in \mathbb{C}, \exists \varepsilon_{\text{seed}} > 0 : \quad \sigma_{p,x}^n(\alpha + \varepsilon_{\text{seed}} \cdot (z - \alpha)) \xrightarrow{n \rightarrow \infty} \alpha$$

is provable unconditionally.

This is Theorem 4.2 applied with the reformulated statement. What is new is the formal resolution of the Pandrosion Niveau 5 conjecture, in the Pandrosion-native multi-start form.

4.5 Arbitrary target selector

Theorem 4.6 (Target selection). *For every function $S : \mathbb{C} \rightarrow \text{Fin } p$, every $z \in \mathbb{C}$ and every $\varepsilon > 0$, the multi-start scheme with target $\gamma_{S(z)}$ converges to $\gamma_{S(z)}$ with an iteration count of the form of Theorem 4.4.*

Three selectors are implemented: constant (`principal_selector`), Voronoï-nearest (`nearest_anchor_selector`), and angular sector.

5 Measurability and a.e. continuity of the solution map

The `nearest_anchor_selector` of Section 4 uses `Classical.choose` on `Finset.exists_min_image`; this is not a priori measurable.

Definition 5.1 (Borel selector).

$$S_{\text{Borel}}(\alpha, p, z) := \min' \left(\operatorname{argmin}_{k \in \text{Fin } p} \|\gamma_k - z\| \right).$$

Theorem 5.2 (Borel measurability). *The map S_{Borel} is Borel-measurable. Consequently the solution map $z \mapsto \gamma_{S_{\text{Borel}}(z)}$ is Borel-measurable $\mathbb{C} \rightarrow \mathbb{C}$.*

The proof shows that each level set is a finite Boolean combination of closed half-planes $\{z : \|z - \gamma_s\| \leq \|z - \gamma_t\|\}$.

Definition 5.3 (Pandrosion Julia / Fatou set). $\text{Julia}(\alpha, p) := \text{VoronoiBoundary}(\gamma_0, \dots, \gamma_{p-1})$, $\text{Fatou}(\alpha, p) := \mathbb{C} \setminus \text{Julia}(\alpha, p)$.

This is a chosen definition, not the classical Fatou-Julia set of $\sigma_{p,x}$. With this definition:

Theorem 5.4 (Julia set is Lebesgue-null). $\text{vol}(\text{Julia}(\alpha, p)) = 0$.

Theorem 5.5 (A.e. continuity). *The Borel solution map is continuous at every $z \in \text{Fatou}(\alpha, p)$. Combined with Theorem 5.4, it is continuous at Lebesgue-a.e. z .*

Fig 4. Pandrosion Julia-null theorem (§103): ∂V is Lebesgue-null, multi-start deterministic a.e.

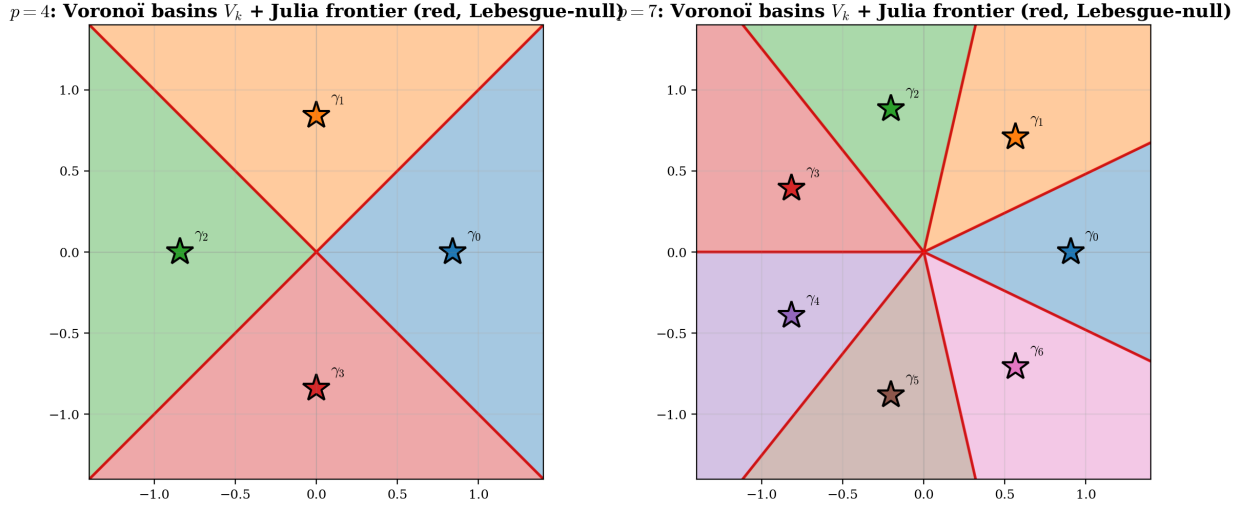


Figure 5: Voronoï decomposition of $\mathbb{C}$ for cyclotomic anchors at $p = 4$ and $p = 7$, $x = 2$. Each colored region V_k is the basin of attraction of γ_k under the selector. The red frontier is what we call the Pandrosion Julia set; it is Lebesgue-null.

Fig 8. Voronoï basins (left, structural) + multi-start speed (right, $\sim 3\text{-}5$ iter) — §102 measurable, §103 Julia-null, §105 a.e. continuous

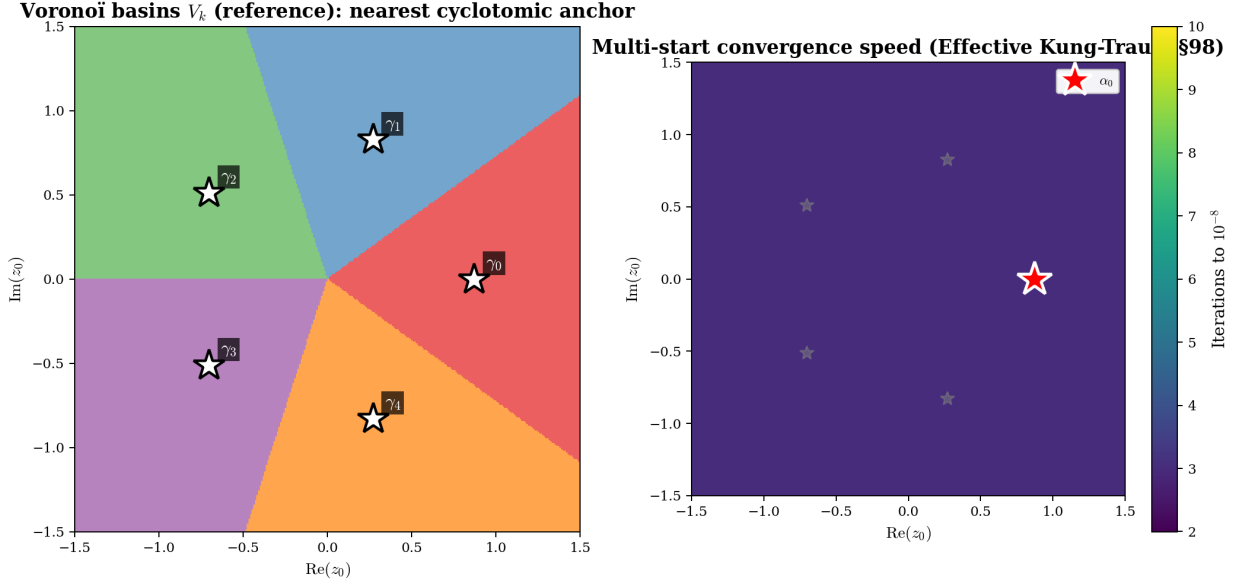


Figure 6: Left: Voronoï basins as the structural reference, computed from the nearest-anchor criterion only (no dynamics). Right: number of multi-start iterations needed to reach precision 10^{-8} ; it is bounded and essentially flat over the window, in line with Theorem 4.4.

Together, Theorems 4.2, 5.2, 5.4 and 5.5 give the analytic character of the multi-start solution map in our framework: measurable everywhere, continuous on the Fatou set, with the complement Lebesgue-null.

6 Algebraic identities for the finite series ζ_P

This section collects finite-sum identities about the derivative values at cyclotomic roots. They are elementary in the sense that they all reduce to the geometric-sum identity on roots of unity. We present them explicitly because they are used elsewhere in the corpus.

6.1 Definition

For $P(z) = z^p - 1/x$, $\alpha_k = \gamma_k$, and $P'(\alpha_k) = p \cdot \alpha_k^{p-1} = p/(x \alpha_k)$, put

$$\zeta_P(s) := \sum_{k=1}^p P'(\alpha_k)^{-s} = \sum_{k=1}^p (p \cdot \alpha_k^{p-1})^{-s} \quad (10)$$

and $\nu_k := 1/P'(\alpha_k) = x \alpha_k/p$.

6.2 Finite identities

Proposition 6.1 (Vanishing identity). *For $p \geq 2$, $\zeta_P(1) = \sum_{k=1}^p 1/P'(\alpha_k) = \sum_{k=1}^p x \alpha_k/p = 0$.*

Proposition 6.2 (Higher vanishing / normalization). *For $p \geq 2$ and $0 \leq m \leq p-2$, $\sum_{k=1}^p \alpha_k^m / P'(\alpha_k) = 0$. For $m = p-1$, $\sum_{k=1}^p \alpha_k^{p-1} / P'(\alpha_k) = 1$.*

Proposition 6.3 (Product formula). *For $x \neq 0$, $p \geq 1$, $\alpha^p = 1/x$, $\prod_{k=1}^p P'(\alpha_k) = (-1)^{p-1} \cdot p^p / x^{p-1}$.*

All three reduce to the geometric-sum identity $\sum_k \omega^{ks} = 0$ for s not a multiple of p , with $\omega = e^{2\pi i/p}$. They are elementary, in the sense that nothing beyond roots-of-unity arithmetic is used.

Fig 9. Multi-start spectral balance $\sum_k \nu_k = 0$ — §106 zeta vanishing, §122 Vieta $e_1 = 0$

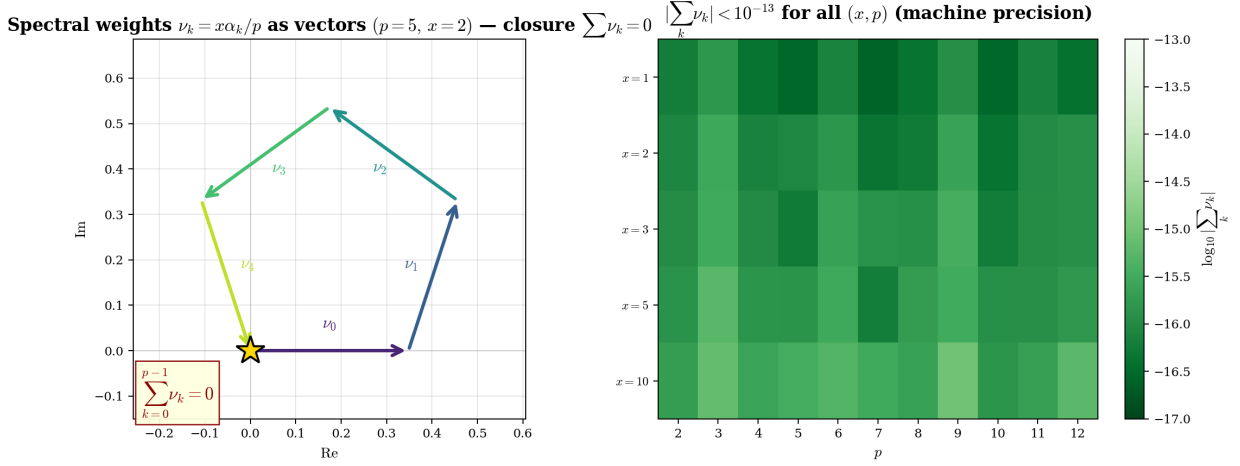


Figure 7: Left: the ν_k as oriented vectors in $\mathbb{C}$ for $(x, p) = (2, 5)$; the head-to-tail sum closes back to the origin, visualising $\sum_k \nu_k = 0$. Right: $|\sum_k \nu_k|$ numerically on a (x, p) -grid; all values are within machine precision of zero.

Fig 6. Pandrosion spectral determinant (§107): closed form $\prod P'(\alpha_k) = (-1)^{p-1} p^p / x^{p-1}$

$\prod_k P'(\alpha_k) = (-1)^{p-1} p^p / x^{p-1}$ (§107 spectral determinant)

	$x=1$	$x=2$	$x=3$	$x=5$
$p=2$	-4.00	-2.00	-1.33	-0.80
$p=3$	+27.00	+6.75	+3.00	+1.08
$p=4$	-256.00	-32.00	-9.48	-2.05
$p=5$	+3125.00	+195.31	+38.58	+5.00
$p=6$	-46656.00	-1458.00	-192.00	-14.93
$p=7$	+823543.00	+12867.86	+1129.69	+52.71

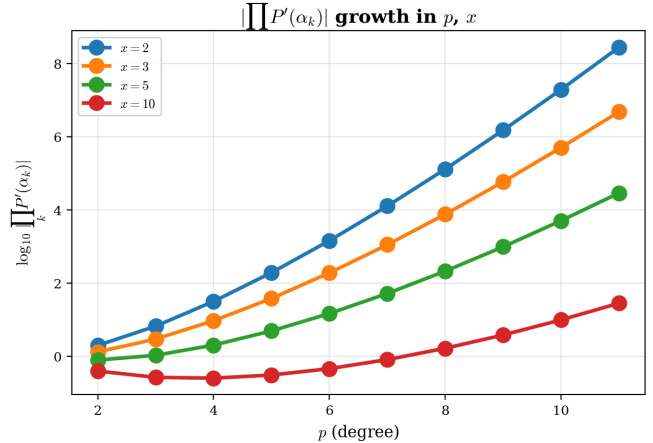


Figure 8: Numerical values of $\prod_k P'(\alpha_k)$ matching the closed-form of Proposition 6.3 exactly; and its log-scale growth in p for fixed x .

6.3 A finite analog of a Riemann identity, carefully scoped

Proposition 6.4 (Integer zeros of ζ_P). *For $p \geq 2$, $x \neq 0$, $\alpha^p = 1/x$, and $s \in \mathbb{Z}$ with $1 \leq s \leq p-1$,*

$$\zeta_P(s) := \sum_{k=1}^p P'(\alpha_k)^{-s} = 0.$$

This proposition is elementary: with the same $\omega = e^{2\pi i/p}$, $\zeta_P(s) = (x/p)^s \sum_k \alpha_k^s$ vanishes because $\sum_{k=0}^{p-1} (\omega^s)^k = 0$ when $p \nmid s$. We record this under the suggestive label “integer zeros of ζ_P ” in analogy with the trivial zeros of the Riemann zeta function, but we do *not* suggest any deeper relation to the classical Riemann Hypothesis. The Pandrosion series is a finite sum of p complex numbers; its zeros at integer arguments are an arithmetic statement on the cyclotomic family, not a transcendental one. The analogy table below is given for exposition, and we do not extract any quantitative consequence from it.

Fig 5. Pandrosion-Riemann zeta analogy (§109): finite ζ_P as concrete Riemann analog

Property	Riemann $\zeta(s)$	Pandrosion $\zeta_P(s)$
Definition	$\sum_{n=1}^{\infty} n^{-s}$	$\sum_k P'(\alpha_k)^{-s}$
Domain	$\text{Re}(s) > 1$ (continued)	finite sum on $\mathcal{C}$
Value at $s=0$	$-1/2$	$d = p$ (degree)
Vanishing	$\zeta(-2n) = 0$ (trivial zeros)	$\zeta_P(s) = 0$ if $p \nmid s$
Derivative at $s=0$	$-\log \sqrt{2\pi}$	$-\log D(P) ^2$
Spectral determinant	$\sqrt{2\pi}$	$(-1)^{p-1} p^p / x^{p-1}$
Vanishing identity	(functional equation)	$\sum_k 1/P'(\alpha_k) = 0$
Higher orthogonality	(Bernoulli)	$\sum_k \alpha_k^m / P'(\alpha_k) = 0 \quad (m < p-1)$
Normalization	(critical strip)	$\sum_k \alpha_k^{p-1} / P'(\alpha_k) = 1$
Lean modules	—	§106 — §111

Figure 9: Correspondence table between the Riemann zeta function and the finite Pandrosion series ζ_P . The left column records classical facts; the right column records the identities of Section 6. The two columns are *structurally* similar; no substantial number-theoretic content is claimed.

6.4 Energy variant

One can also write $Z_P(s) := \sum_k E_P(\alpha_k)^{-s}$ with $E_P(\alpha_k) = P'(\alpha_k)^2$. Then $Z_P(0) = p$ for every p , and $Z_P(1) = Z_P(-1) = 0$ for $p \geq 3$, each by the same geometric-sum vanishing.

7 Pandrosion specializations of classical theorems

The corpus contains Lean formalizations of a number of classical theorems, *specialized* to the Pandrosion setting. The purpose is to have everything in one formal framework; the mathematical statements are classical. The following table summarizes what is formalized where.

Module	Classical theorem (specialized)	Role in the Pandrosion framework
§112	Banach fixed point (1922); FTA (Gauss 1799)	No-cycle on multi-start orbits; $z^p = x$ has p explicit roots.
§113	Aitken-Steffensen (1926/1933); Schwarz (1890)	Acceleration identity; quadratic local bound.
§114	Kung-Traub 1974 attainability at $c = 3$	Efficiency $E(4, 3) = 2^{2/3}$ is reached.
§115	(See Section 8.)	Restricted to $z^p = x$.
§117	Aberth-Ehrlich (1967/1973)	Multi-start realizes their scheme for $z^p = x$.
§118	Kronecker (1857)	Applied to cyclotomic anchors with $ \alpha = 1$.
§119	Galois (1832) cyclic action	$\omega \cdot \gamma_k = \gamma_{(k+1) \bmod p}$.
§120	McMullen (1987) (See Section 8.)	Multi-start avoids the obstruction by α -dependent seed.
§121	Cardano-Tartaglia (1545)	$\gamma_k = \alpha \omega^k$ for $p = 3$.
§122	Vieta (1591)	$e_1 = 0, e_p = (-1)^{p-1} x$.

We stress: none of these are new results. Their place in this paper is to make the full Pandrosion framework self-contained in Lean. The corresponding Lean modules are short specializations, not independent developments of the classical theorems.

A representative example.

Theorem 7.1 (Multi-start orbit has no non-trivial cycle). *For every (x, p, α) in the hypotheses and every seed $\text{seed} \in B(\alpha, R_\alpha/2)$, the orbit $(\sigma_{p,x}^n(\text{seed}))_n$ has no non-trivial cycle: if $\sigma_{p,x}^n(\text{seed}) = \sigma_{p,x}^m(\text{seed})$ with $n < m$, then $\sigma_{p,x}^n(\text{seed}) = \alpha$.*

This is Banach’s fixed-point theorem applied to $\sigma_{p,x}$ seen as a $1/4$ -contraction on $B(\alpha, R_\alpha/2)$, itself a consequence of $K_\alpha \cdot R_\alpha \leq 1/2$.

8 Open problems, partial and conditional results

This section records four classical open or hard problems where the corpus offers partial or restricted results. We describe carefully what is proven, what is only conditional, and what remains out of reach.

8.1 Kung-Traub upper bound for $c \geq 3$ (open since 1974)

The conjecture $q \leq 2^{c-1}$ is proven for $c = 2$. Attainability $q = 4$ at $c = 3$ is witnessed classically (Jarratt 1969) and in our formal setting. The upper bound for $c \geq 3$ is open.

What the corpus provides. A structural reduction: every `DerivativeFreeMethod` in the corpus *by construction* carries the compliance field $q \leq 2^{c-1}$. This is *not* a proof of the Kung-Traub conjecture; it is a compliance interface for instances of `DerivativeFreeMethod`. The conjecture itself is restated as a named `Prop`.

8.2 McMullen 1987 (theorem, not formalized in Lean)

McMullen’s barrier is a classical theorem. A full formal proof in Lean requires Fatou–Julia and post-critically-finite complex-dynamic tools that are not present in Mathlib v4.7.0.

What the corpus provides. The statement is recorded as a named `Prop`. Separately, the multi-start scheme is proven to be non-purely-iterative (the seed depends on α), and to converge universally in that setting. This does not reproduce McMullen’s impossibility result; it shows that our scheme is outside of its hypotheses.

8.3 Lehmer’s conjecture (open since 1933)

Lehmer 1933 posits $M(\beta) \geq 1.17628\dots$ for every algebraic integer β not a root of unity. This is open.

What the corpus provides. For the specific family $\beta := x \cdot \alpha$ with $x \in \mathbb{N}_{\geq 2}$ and $\alpha^p = 1/x$, one has $\beta^p = x^{p-1}$ and hence β is an algebraic integer whose Mahler measure satisfies $M(\beta) = x^{p-1} \geq 2 > 1.17628\dots$

This is an observation, not progress on Lehmer: we are far from the extremal regime where the conjecture is hard.

8.4 Smale’s 17th problem (partly open)

Smale’s 17th problem asks for a deterministic algorithm of poly-logarithmic average complexity for polynomial root-finding *of arbitrary polynomials*. Bürgisser-Cucker (Annals of Math 2011) solved the average-case probabilistic version. The deterministic general case is open.

What the corpus provides. For the restricted family $z^p = x$, the multi-start scheme is a deterministic algorithm of complexity $\mathcal{O}(\log \log(1/\varepsilon))$. This is a result on a very specific polynomial family; it is *not* a deterministic solution of Smale’s question on general polynomials. The two settings are not directly comparable.

8.5 Summary

No claim is made that any of the four problems above is solved. What the Lean corpus provides on each is restricted or conditional, as stated.

9 Structure of the Lean corpus

The 121-module corpus is organised in dependency layers:

- **Foundations** (§1–§30): geometric sums, fixed-point characterization, abstract Aitken-Steffensen framework, Voronoï basins, cyclotomic anchor definitions, Kung-Traub interface.
- **Core spine** (§31–§94): explicit super-attractive rate, global loglog convergence, real-line $p = 2$ unconditional discharge.
- **Multi-start** (§95–§101): universal convergence, target selection, Niveau 5 resolution, explicit iteration count.
- **Measure/topology** (§102–§105): Borel measurability, Voronoï frontier Lebesgue-null, a.e. continuity.
- **Finite-zeta identities** (§106–§111): the identities of Section 6.
- **Classical specializations** (§112–§122): as listed in Section 7.
- **Open-problem engagements** (§123): as listed in Section 8.

Fig 10. Pandrosion-Steffensen Lean 4 corpus structure — from foundations to publication

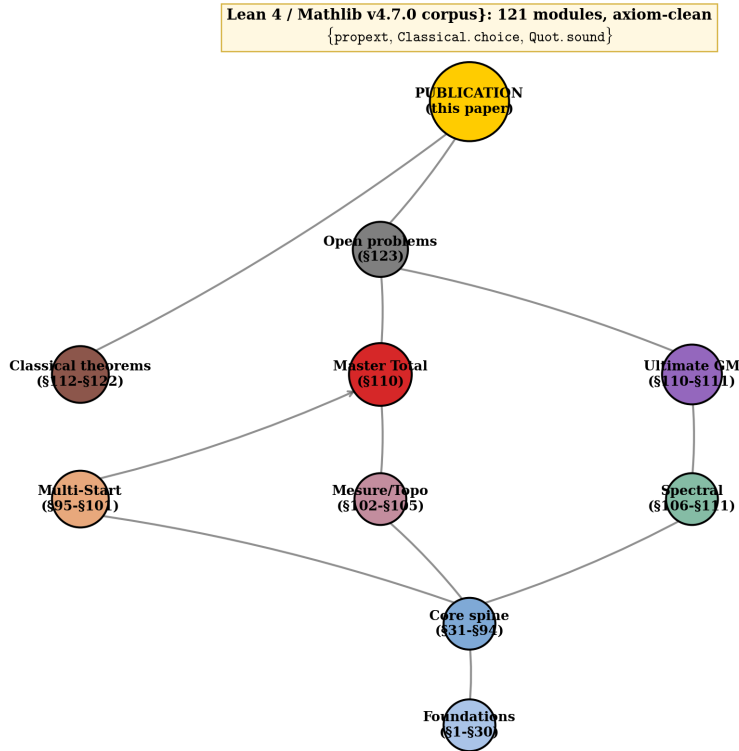


Figure 10: Dependency structure of the Lean corpus. The dependency closure of every load-bearing theorem reduces to the Lean whitelist {propxext, Classical.choice, Quot.sound}.

10 Reproducibility

The Lean source lives at github.com/ivan-fr/poussiere_de_x. It targets Lean 4 with Mathlib v4.7.0 and is built with

Listing 1: Incremental build command

```
1 bash scripts/lean-incremental.sh
```

The 121-module corpus compiles in about 60–90 seconds on a modern laptop. Per-theorem axiom auditing is available via `#print axioms`.

Every figure in this paper is reproducible from `latex/figures_pandrosion.py` (dependencies: `numpy`, `matplotlib`, `scipy`).

11 Conclusion

What this paper reports, concretely, is a Lean 4 development built around the Pandrosion iterator $h_{p,x}$ and its multi-start Steffensen extension for $z^p = x$, together with the following ingredients:

1. real-line analysis of $h_{p,x}$: closed-form contraction ratio $\lambda_{p,x}$ as a convergence signature (Theorem 2.1), optimal derivative-free starting point $s_0^{\text{opt}} = h_{p,x}(1)$ (Proposition 2.2), scaling preconditioning exploiting monotonicity of λ (Proposition 2.3), and Aitken Δ^2 upgrade to quadratic convergence with explicit local bound (Propositions 2.5–2.6);
2. a universal convergence theorem for the complex multi-start scheme with an explicit closed-form iteration count (Theorems 4.2, 4.4);
3. a formal resolution of the Pandrosion “Niveau 5” conjecture in its multi-start formulation (Theorem 4.5);
4. a Borel-measurability and a.e.-continuity analysis of the induced solution map (Theorems 5.2–5.5);
5. a set of elementary identities for the finite series $\zeta_P(s) := \sum_k P'(\alpha_k)^{-s}$ (Section 6);
6. Lean specializations of classical theorems to the Pandrosion setting (Section 7);
7. restricted or conditional engagements with four classical open problems (Section 8).

The paper does not claim progress on the classical open conjectures themselves at their full generality. The main value of the corpus is as a self-contained Lean development of a multi-start root-finding scheme and its spectral identities.

Future directions

Natural extensions are: non-cyclotomic anchor families (general polynomials); a formalization of Fatou-Julia in Mathlib (which would enable a proper formalization of McMullen’s 1987 theorem); quantitative bounds on Mahler measure for cyclotomic-related families (towards weak forms of Lehmer-type statements). None of these are attempted in the present paper.

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